MATHEMATICS

BINOMIAL THEOREM

THEORY SHEET

ENGLISH

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1.1 INTRODUCTION

Binomial Expression : An expression containing two terms

For example $a + b/x$, $x + 1/y$, $a - y^2$ etc. are binomial expressions.

Trinomial Expression : Expression containing three terms For example $x + y + z$

Multinomial Expression : An expression containing more than two terms.

Pascal's triangle : A triangular arrangement of numbers as shown. The numbers give the coefficients for the expansion of $(x + y)^n$. The first row is for n = 0, the second for n = 1, etc. Each row has 1 as its first and last number. Other numbers are generated by adding the two numbers immediately to the left and right in the row above.

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
etc.

1.2 NEWTON'S BINOMIAL THEOREM

If $x, y \in R$ and $n \in N$, then :

$(x + y)^n = {}^nC_0x^n + {}^nC_1x^{n-1}y + {}^nC_2x^{n-2}y^2 + ... + {}^nC_rx^{n-r}y^r + ... + {}^nC_ny^n$

Q.1 Expand $(x + 2y)^5$

Ans $x^5 + 10x^4y + 40x^3y^2 + 80x^2y^3 + 80xy^4 + 32y^5$

Sol. $= {}^5C_0\,x^5(2y)^0 + {}^5C_1x^4(2y)^1 + {}^5C_2x^3(2y)^2 + {}^5C_3x^2(2y)^3 + {}^5C_4x^1(2y)^4 + {}^5C_5x^0(2y)^5$

$= x^5 + 10x^4y + 40x^3y^2 + 80x^2y^3 + 80xy^4 + 32y^5$

Q.2 Expand $(1 + x)^n$

Ans $(1 + x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + ... + {}^nC_nx^n.$

Sol. $(1 + x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + {}^nC_4\,x^4 + ... + {}^nC_nx^n.$

Q.3 Expand $(1 - x)^n$

Ans $(1 - x)^n = {}^nC_0 - {}^nC_1x + {}^nC_2x^2 + ... + (-1)^n\,.\,{}^nC_nx^n.$

Sol. $(1 - x)^n = {}^nC_0 - {}^nC_1x + {}^nC_2x^2 - {}^nC_3x^3 + {}^nC_4\,x^4 - {}^nC_5\,x^5 + ... + (-1)^n\,.\,{}^nC_nx^n.$

Q.4 Find value of $(2 + \sqrt{3})^5 + (2 - \sqrt{3})^5$

Ans 724

Sol. $\because \left(2+\sqrt{3}\right)^5 = {}^5C_0 2^5 + {}^5C_1 2^4\left(\sqrt{3}\right)^1 + {}^5C_2 2^3\left(\sqrt{3}\right)^2 + {}^5C_3 2^2\left(\sqrt{3}\right)^3 + {}^5C_4 2^1\left(\sqrt{3}\right)^4 + {}^5c_5\left(\sqrt{3}\right)^5$ .....(1)

and $\left(2-\sqrt{3}\right)^5 = {}^5C_0 2^5 - {}^5C_1 2^4\left(\sqrt{3}\right) + {}^5C_2 2^3\left(\sqrt{3}\right)^2 - {}^5C_3(2)^2\left(\sqrt{3}\right)^3 + {}^5C_4 2^1\left(\sqrt{3}\right)^4 - {}^5C_5\left(\sqrt{3}\right)^5$ .....(2)

Now equation (1) + (2) :

$\left(2+\sqrt{3}\right)^5 + \left(2-\sqrt{3}\right)^5 = 2\left[{}^5C_0 2^5 + {}^5C_2 2^3(3) + {}^5C_4 2^1(3)^2\right]$

$= 2[32 + 240 + 90]$

$= 2 \times 362 = 724$

Q.5 Find value of $(\sqrt{2}+1)^6-(\sqrt{2}-1)^6$

Ans. $140\sqrt{2}$

Sol. $\because \left(\sqrt{2}+1\right)^6 = {}^6C_0\left(\sqrt{2}\right)^6 + {}^6C_4\left(\sqrt{2}\right)^5 + {}^6C_2\left(\sqrt{2}\right)^4 + {}^6C_3\left(\sqrt{2}\right)^3 + {}^6C_4\left(\sqrt{2}\right)^2 + {}^6C_3\left(\sqrt{2}\right)^1 + {}^6C_6$ .....(1)

and $\left(\sqrt{2}-1\right)^6 = {}^6C_0\left(\sqrt{2}\right)^6 - {}^6C_1\left(\sqrt{2}\right)^5 + {}^6C_2\left(\sqrt{2}\right)^4 - {}^6C_3\left(\sqrt{2}\right)^3 + {}^6C_4\left(\sqrt{2}\right)^2 - {}^6C_5\left(\sqrt{2}\right)^1 + {}^6C_6$ .....(2)

eq (1) – (2) :

$$\left(\sqrt{2}+1\right)^6 - \left(\sqrt{2}-1\right)^6 = 2\left[{}^6C_1\left(\sqrt{2}\right)^5 + {}^6C_3\left(\sqrt{2}\right)^3 + {}^6C_5\left(\sqrt{2}\right)^1\right]$$

$$= 2\left[24\sqrt{2} + 40\sqrt{2} + 6\sqrt{2}\right]$$

$$= 2\sqrt{2}\times 70 = 140\sqrt{2}$$

1.3 HIGHLIGHTS OF BINOMIAL THEOREM

In the expansion of $(x+y)^n$

1. The number of terms in the expansion is (n+1) i.e. one more than the index.
2. The sum of the indices of x & y in each term is n.
3. ${}^nC_0, {}^nC_1, {}^nC_2, \ldots\ldots\ldots, {}^nC_n$ are called binomial coefficient.
4. Difference between coefficient and binomial coefficient.

Expansion	$(x-2y)^5 =$	${}^5C_0x^5$	$-{}^5C_1x^4(2y) + {}^5C_2x^3(2y)^2 + \ldots\ldots$	
Binomial coefficient		${}^5C_0$	${}^5C_1$	${}^5C_2$
Coefficient		${}^5C_0$	$-{}^5C_1(2)$	${}^5C_2(2)^2$

Sum of binomial coefficients $= {}^nC_0 + {}^nC_1 + {}^nC_2 + \ldots\ldots + {}^nC_n = 2^n$

5. To find sum of coefficients, put variable = 1

Example	Expansion	Sum of binomial coefficient	Sum of coefficient
	$(2x+y)^6$	2^6	3^6
	$(x-2y)^5$	2^5	-1
	$(x-y)^7$	2^7	0

6. The binomial coefficients of the terms $({}^nC_0, {}^nC_1\ldots\ldots)$ equidistant from the beginning and the end are equal. i.e. ${}^nC_r = {}^nC_{r-1}$
7. Some important expansions :

(i) $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \ldots\ldots\ldots + {}^nC_nx^n$.

(ii) $(1-x)^n = {}^nC_0 - {}^nC_1x + {}^nC_2x^2 + \ldots\ldots\ldots + (-1)^n \cdot {}^nC_nx^n$.

8. The coefficient of x^r in $(1+x)^n = {}^nC_r$

9. $(x+y)^n = \sum_{r=0}^{n} {}^nC_r x^{n-r}y^r$

2.1 GENERAL TERM

$(x + y)^n = {}^nC_0\, x^n\, y^0 + {}^nC_1\, x^{n-1}\, y^1 + \ldots\ldots\ldots + {}^nC_r\, x^{n-r}\, y^r + \ldots\ldots\ldots + {}^nC_n\, x^0\, y^n$

$(r + 1)^{th}$ term is called general term and denoted by T_{r+1}.

$$T_{r+1} = {}^nC_r\, x^{n-r}\, y^r$$

Q.1 The tenth term in the expansion of $(2x^2 + 1/x)^{12}$ is

Ans $1760/x^3$

Sol. Comparing $(2x^2 + 1/x)^{12}$ with $(X + a)^n$.

$n = 12,\ X = 2x^2,\ a = 1/x.$

$\therefore\ 10^{th}$ term $= T_{10} = {}^{12}C_9(2x^2)^{12-9}(1/x)^9$

$= {}^{12}C_9.8.1/x^3$

or $T_{10} = 1760/x^3$

Q.2 The 4th term from the end in the expansion of $(2x - 1/x^2)^{10}$ is

Ans $-960\ x^{-11}$

Sol. Required term $= T_{10-4+2} = T_8$

$= {}^{10}C_7(2x)^3(-1/x^2)^7$

$= -960\ x^{-11}$

Q.3 In the expansion of $\left(x^2 + \frac{2}{x}\right)^6$, find.

(i) Term involving x^3. (ii) Coefficient of x^3.

(iii) Term indendent of x (iv) Coefficient of x^2.

Ans. (i) $160\ x^3$ (ii) 160 (iii) 240 (iv) 0

Sol. $\because$ General term is given by

$T_{r+1} = {}^nc_r(a)^{n-r}(b)^r$

$T_{r+1} = {}^6C_r\left(x^2\right)^{6-r} \cdot \left(\frac{2}{x}\right)^r$

$T_{r+1} = {}^6C_r x^{12-2r} \cdot 2^r \cdot x^{-r}$

$T_{r+1} = {}^6C_r 2^r \cdot x^{12-3r}$

(1) for term in volving x^3 put $12 - 3r = 3$

$\Rightarrow r = 3$

$\therefore\ T_4 = {}^6C_3\, 2^3\, .\, x^3$

$= 20 \times 8x^3 = 160x^3.$

(2) Clearly coeff of $x^3 = 160$

(3) For term independent of x, put $12 - 3r = 0$

$\Rightarrow r = 4$

$\therefore$ Term independent of $x = P_5 = {}^6C_4.2^4$

$= 240$

(4) For coeff. of x^2 put $12 - 3r = 2$

$\Rightarrow r = \frac{10}{3}$ is not an integer.

$\therefore$ Coeff. of $x^2 = 0$

Q.4 Find number of rational terms in the expansions of $\left(5^{1/3} + 7^{1/4}\right)^{60}$.

Ans. 6

Sol. $\because T_{r+1} = {}^{60}C_r \left(5^{\frac{1}{3}}\right)^{60-r} \cdot \left(7^{\frac{1}{4}}\right)^r$

$= {}^{60}C_r \, 5^{\frac{60-r}{3}} \cdot 7^{\frac{r}{4}}$

Clearly for rational terms

$\frac{60-r}{3} \in z$ and $\frac{r}{4} \in z$ and $0 \le r \le 60$

$r = 0, 4, 8, 12, 16, 20, ..., 60$

$r = 0, 12, 24, 36, 48, 60$ (6 values)

$\therefore$ Total no. of rational terms = 6

Q.5 Find number of rational terms in the expansions of $\left(5^{1/2} + 7^{1/4}\right)^{99}$

Ans. 0

Sol. $\because$ General term is :

$T_{r+1} = {}^{99}C_r \left(5^{\frac{1}{2}}\right)^{99-r} \left(7^{\frac{1}{4}}\right)^r$

$T_{r+1} = {}^{99}C_r \, 5^{\frac{99-r}{2}} \cdot 7^{r/4}$

Clearly for rational term :

$\frac{99-r}{2} \in \cdot z$ and $\frac{r}{4} \in z$ and $0 \le r \le 99$

$r = 0, 4, 8, 12, 16, 20, ..., 96$

$r \in \phi$

$\therefore$ Total number of rational terms = 0

2.2 MIDDLE TERM

Middle term has greatest binomial coefficient and if there are 2 middle terms their coefficients will be equal.

Q.1 Find position of middle term for following expansions.

(i) $(x+y)^{30}$ (ii) $(x+y)^{49}$

Ans. (i) 16th term (ii) 25th term & 26th term

Sol. (i) $\because$ $n = 30$ (even) $\Rightarrow$ Only one middle term.

$$\therefore r = \frac{30}{2} = 15$$

So middle terms $= T_{r+1} = T_{15+1} = T_{16}$

i.e. 16th term is middle term

(ii) $\because$ $n = 49$ (odd) $\Rightarrow$ two middle term

for $r = \frac{n-1}{2}$ and $r = \frac{n+1}{2}$

$$r = \frac{49-1}{2} = 24 \text{ and } r = \frac{49+1}{2} = 25$$

$\Rightarrow T_{r+1} = T_{25}$ $\Rightarrow T_{r+1} = T_{26}$

i.e. 25th and 26th term is middle term.

Q.2 Find coefficient of x^3 in expansion of $(2+3x-5x^2)(1+x)^{10}$

Ans. 325

Sol. $\because$ General term in the expansion of $(1+x)^{10}$ is given by $T_{r+1} = {}^{10}C_r x^r$

$\therefore$ Coeff. of x^3 in the exponsion of $(2+3x-5x^2)(1+x)^{10}$ is given by

$= 2 \cdot {}^{10}C_3 + 3 \cdot {}^{10}C_2 - 5 \cdot {}^{10}C_1$

$= 2 \times 120 + 3 \times 45 - 5 \times 10$

$= 240 + 135 - 50 = 325$

Q.3 Find coefficient of x^3 in expansion of $\left(5+4x+\frac{3}{x}\right)(1+x)^{10}$

Ans. 990

Sol. $\left(5+4x+\frac{3}{x}\right)\left({}^{10}C_r x^r\right)$

$\therefore$ Coeff. of $x^3 = 5 \cdot {}^{10}C_3 + 4 \cdot {}^{10}C_2 + 3 \cdot {}^{10}C_4$

$= 5 \times 120 + 4 \times 45 + 210$

$= 600 + 180 + 210$

$= 990$

Q.4 Find coefficient of x^4 $(1+x^2)^6 (1+x)^{10}$

Ans. 495

Sol. $\left({}^{6}C_{r}\left(x^{2}\right)^{r}\right)\cdot\left({}^{10}C_{r}x^{r}\right)$

$\therefore$ Coeff. of $x^4 = {}^{6}C_{0}\ .\ {}^{10}C_{4} + {}^{6}C_{1}\ {}^{10}C_{2} + {}^{6}C_{2}\ {}^{10}C_{4}$

$= 1.210 + 6 \times 45 + 15 \times 1$

$= 210 + 270 + 15 = 495$

Q.5 Find coefficient of x^3 in expansion of $\left(2+4x+5x^{2}\right)\left(2x^{2}-\frac{3}{x}\right)^{6}$.

Ans. 8640

Sol. $\left(2+4x+5x^{2}\right)\times {}^{6}C_{r}\left(2x^{2}\right)^{6-r}\left(-\frac{3}{x}\right)^{r}$

$=\left(2+4x+5x^{2}\right)\times {}^{6}C_{r}\,2^{6-r}\cdot(-3)^{r}x^{12-3r}$

$= 2 \times \text{coeff. of } x^{3}\left(\begin{array}{c}\text{put } 12-3r=3\\ r=\frac{9}{3}=3\end{array}\right)$

$+ 4 \times \text{coeff. of } x^{2}\left(\begin{array}{c}\text{put } 12-3r=2\\ r=\frac{10}{3}\notin z\end{array}\right)$

$+ 5 \times \text{coeff of } x^{1}\left(\begin{array}{c}\text{put } 12-3r=1\\ r=\frac{11}{3}\notin z\end{array}\right)$

$= 2 \times {}^{6}C_{3}\ 2^{3}\ (-3)^{3} + 0 + 0.$

$= 2 \times 20 \times 8 \times (-27) = -320 \times 27$

$= -8640$

3. NUMERICALLY GREATEST TERM

In expansion of $(x + y)^n$

$|T_{r+1}| > |T_r| \quad \Rightarrow \left|\frac{T_{r+1}}{T_r}\right| > 1 \quad \Rightarrow r < \frac{n+1}{1+\left|\frac{x}{y}\right|} \quad \left(\text{Let : } p = \frac{n+1}{1+\left|\frac{x}{y}\right|}\right)$

	$\lvert T_{r+1}\rvert > \lvert T_r\rvert$	$\lvert T_{r+1}\rvert = \lvert T_r\rvert$	$\lvert T_{r+1}\rvert < \lvert T_r\rvert$
$\Rightarrow$	$r < p$	$r = p$	$r > p$

C - I $p \neq$ Integer Let $p = 4.3$

	$\lvert T_{r+1}\rvert > \lvert T_r\rvert$	$\lvert T_{r+1}\rvert = \lvert T_r\rvert$	$\lvert T_{r+1}\rvert < \lvert T_r\rvert$

$r < 4.3$ $\quad r = 4.3$ $\quad r > 4.3$

$|T_8| < |T_7| < |T_6| < |T_5| > |T_4| > |T_3| > |T_2| > |T_1|$

T_5 is NGT $\Rightarrow$ $T_{[p]+1}$ will be NGT

Case - II $\quad p =$ Integer $\quad$ Let $p = 4$

$|T_{r+1}| > |T_r|$ $\quad |T_{r+1}| = |T_r|$ $\quad |T_{r+1}| < |T_r|$

$r < 4$ $\quad r = 4$ $\quad r > 4$

$|T_8| < |T_7| < |T_6| < |T_5| = |T_4| > |T_3| > |T_2| > |T_1|$

T_4 and T_5 are NGT $\Rightarrow$ T_p and T_{p+1} are NGT

Q.1 In the expansion of $(3-2x)^9$ find position of NGT if

(i) $x = \frac{1}{2}$ $\quad$ (ii) $x = 1$ $\quad$ (iii) $x = \frac{3}{2}$

Ans. (i) T_3 $\quad$ (ii) T_4 & T_5 $\quad$ (iii) T_5 & T_6

Sol. For NGT : $\frac{n+1}{\left|\frac{a}{b}\right|+1} - 1 \le T_r \le \frac{n+1}{\left|\frac{a}{b}\right|+1}$

$\because a = 3, \quad b = -2x, \quad n = 9$

$\therefore \quad \frac{10}{\left|\frac{3}{2x}\right|+1} - 1 \le r \le \frac{10}{\left|\frac{3}{2x}\right|+1}$ .....(1)

(1) put $x = \frac{1}{2}$ in (1) :-

$\frac{10}{3+1} - 1 \le T_r \le \frac{10}{3+1}$

$\Rightarrow \frac{3}{2} \le r \le \frac{5}{2}$ and $r \in Z$

$\Rightarrow r = 2$

$\therefore$ NGT $= T_{r+1} = T_{2+1} = T_3$

(ii) Put $x = 1$ in equation 1 :-

$\frac{10}{\frac{3}{2}+1} - 1 \le r \le \frac{10}{\frac{3}{2}+1}$

$3 \le r \le 4$ and $r \in z$

$\Rightarrow r = 3$ and $r = 4$.

$\therefore$ NGT $= T_{r+1} = T_{3+1}$ and T_{4+1}

$= T_4$ and T_5

(iii) Put x = 3/2 in equation 1 :-

$$\frac{10}{1+1}-1 \le r \le \frac{10}{1+1}$$

$\Rightarrow 4 \le r \le 5$ and $r \in Z$

Hence r = 4 and r = 5

Q.2 In expansion of $\left(2+\frac{3x}{8}\right)^{10}$ if T_4 is NGT then find range of x.

Ans $x \in \left(-\frac{64}{21}, -2\right) \cup \left(2, \frac{64}{21}\right)$

Sol According to the question, $|t_4| > |t_3|$, $|t_4| > |t_5|$.

Now $t_{r+1} = {}^{10}C_r . 2^{10-r} \left(\frac{3x}{8}\right)^r$

$\therefore \quad t_4 = {}^{10}C_3 . 2^7 . \left(\frac{3x}{8}\right)^3$, $t_3 = {}^{10}C_2 . 2^8 . \left(\frac{3x}{8}\right)^2$ and $t_5 = {}^{10}C_4 . 2^6 . \left(\frac{3x}{8}\right)^4$

Now, $|t_4| > |t_3|$

$\Rightarrow {}^{10}C_3 . 2^7 . \left(\frac{3}{8}\right)^3 . |x|^3 > {}^{10}C_2 . 2^8 . \left(\frac{3}{8}\right)^2 . |x|^2 \quad \ldots (i)$

and $|t_4| > |t_5|$

$\Rightarrow {}^{10}C_3 . 2^7 . \left(\frac{3}{8}\right)^3 . |x|^3 > {}^{10}C_4 . 2^6 . \left(\frac{3}{8}\right)^4 . |x|^4 \ldots (ii)$

from (i)

$$\frac{10.9.8}{6} . \frac{3}{8} . |x|^3 > \frac{10.9}{2} . 2 |x|^2$$

or $45|x|^3 > 90|x|^2$

or $|x|^3 - 2|x|^2 > 0$

or $|x| > 2$, as x can not be zero.

from (ii) $\frac{10.9.8}{6} . 2 |x|^3 > \frac{10.9.8.7}{24} . \frac{3}{8} . |x|^4$

or $|x|^3 > \frac{7}{8} . \frac{3}{8} |x|^4$

or $|x|^3 \left\{1 - \frac{21}{64}|x|\right\} > 0 \qquad \Rightarrow 1 - \frac{21}{64}|x| > 0$

$\therefore |x| < \frac{64}{21}$

Thus, we get $|x| > 2$ and $|x| < \frac{64}{21} = 3\frac{1}{21}$. So $x \in \left(-\frac{64}{21}, -2\right) \cup \left(2, \frac{64}{21}\right)$

Q.3 In the expansion of $\left(\frac{x}{3}+\frac{2}{3}\right)^n$, if T_9 has numerically greatest coefficient then find n

Ans 12

Sol. $\because \frac{n+1}{\left|\frac{a}{b}\right|+1}-1<r<\frac{n+1}{\left|\frac{a}{b}\right|+1}$

$\Rightarrow \frac{n+1}{\left|\frac{x}{2}\right|+1}-1<8<\frac{n+1}{\left|\frac{x}{2}\right|+1}$

put $x = 1$

$\Rightarrow \frac{n+1}{3/2}-1<8<\frac{n+1}{3/2}$

$\Rightarrow \frac{2(n+1)-3}{3}<8$ and $12 < n + 1$

$\Rightarrow 2(n+1)<27$ $n > 11$

$\Rightarrow n+1<\frac{27}{2}$

$\Rightarrow n<\frac{25}{2}$

$\Rightarrow 11<n<\frac{25}{2}$

$\Rightarrow 11 < n < 12.5$ and $n \in Z$

$\Rightarrow n = 12$

4.1 REMAINDER BASED PROBLEMS

Q.1 Show that $9^n + 7$ is divisible by 8, where n is a positive integer.

OR

Find remainder when 9^n is divided by 8

Ans. 1

Sol. $9^n + 7 = (1 + 8)^n + 7$

$= {}^nC_0 + {}^nC_1 . 8 + {}^nC_2 . 8^2 + ... + {}^nC_n 8^n + 7.$

$= 8. C_1 + 8^2. C_2 + ... + C_n . 8^n + 8.$

$= 8\lambda$, where λ is a positive integer

Q.2 Find remainder when 7^{50} is divided by 8

Ans. 1

Sol. $7^{50} = (8-1)^{50}$

$= {}^{50}c_0 8^{50} - {}^{50}c_1 8^{49} + {}^{50}c_2 8^{48} - {}^{50}c_3 8^{47} + \cdots + \ldots {}^{50}c_{49}(8)^1 + {}^{50}c_{50}$

$= 8\left[{}^{50}c_0 8^{49} - {}^{50}c_1 8^{48} + {}^{50}c_2 8^{47} - {}^{50}c_3 8^{46} + \cdots - {}^{50}c_{49}\right] + 1$

$= 8\lambda + 1 \quad ; \quad \lambda \in Z$

Clearly remainder = 1

Q.3 Find remainder when 7^{43} is divided by 8.

Ans. 7

Sol. $\because 7^{43} = (8-1)^{43}$

$= {}^{43}C_0 8^{43} - {}^{43}C_1 8^{42} + {}^{43}C_2 8^{41} - {}^{43}C_3 8^{40} + \ldots\ldots + \ldots\ldots + {}^{43}C_{42}(8)^1 - {}^{43}C_{43}$

$= 8\left[{}^{43}C_0 (8)^{42} - {}^{43}C_1 (8)^{41} + {}^{43}C_2 (8)^{40} - \cdots\cdots + {}^{43}C_{42}\right] - 1$

$= 8\lambda - 1 \quad ; \quad \lambda \in Z$

$\therefore$ Remainder = 7

Q.4 Find remainder when 25^{25} is divided by 8

Ans. 1

Sol. $\because 25^{25} = (24+1)^{25}$

$= {}^{25}C_0 (24)^{25} + {}^{25}C_1 (24)^{24} + {}^{25}C_2 (24)^{23} + {}^{25}C_3 (24)^{22} + \cdots\cdots + {}^{25}C_{24}(24) + {}^{25}C_{25}$

$= 24\left[{}^{25}C_0 (24)^{24} + {}^{25}C_4 (24)^{23} + {}^{25}C_2 (24)^{22} + \cdots\cdots + {}^{25}C_{24}\right] + 1$

$= 24\lambda + 1 \quad ; \quad \lambda \in Z$

$\therefore$ Remainder = 1

Q.5 What is the remainder when 5^{99} is divided by 13.

Ans 8

Sol. $5^{99} = 5.5^{98} = 5.(25)^{49} = 5(26-1)^{49}$

$= 5\left[{}^{49}C_0 (26)^{49} - {}^{49}C_1 (26)^{48} + \ldots\ldots\ldots + {}^{49}C_{48}(26)^1 - {}^{49}C_{49}(26)^0\right]$

$= 5\left[{}^{49}C_0 (26)^{49} - {}^{49}C_1 (26)^{48} + \ldots\ldots\ldots + {}^{49}C_{48}(26)^1 - 1\right]$

$= 5\left[{}^{49}C_0 (26)^{49} - {}^{49}C_1 (26)^{48} + \ldots\ldots\ldots + {}^{49}C_{48}(26)^1 - 13\right] + 60$

$= 13(k) + 52 + 8$ (where k is a positive integer)

$= 13(k+4) + 8$

Hence, remainder is 8.

Q.6 Find remainder when 2^{99} is divided by 9

Ans. 8

Sol. $\because 2^{99} = \left(2^3\right)^{33} = 8^{33} = (9-1)^{33}$

$= {}^{33}C_0 (9)^{33} - {}^{33}C_1 (9)^{32} + {}^{33}C_2 (9)^{31} - {}^{33}C_3 (9)^{30} + \cdots\cdots + {}^{33}C_{32}(9)^1 - {}^{33}C_{33}$

$= 9\left[{}^{33}C_0 (9)^{32} - {}^{33}C_1 (9)^{31} + {}^{33}C_2 (9)^{30} - {}^{33}C_3 (9)^{29} + \cdots\cdots + {}^{33}C_{32}\right] - 1$

$= 9\lambda - 1$

$\therefore$ Remainder is $= 9 - 1 = 8$.

Q.7 Find remainder when 2^{98} is divided by 9

Ans. 4

Sol. $2^{98} = \left(2^3\right)^{32} \cdot 2^2 = 4(8)^{32} = 4(9-1)^{32}$

$= 4\left[{}^{32}C_0 9^{32} - {}^{32}C_1 9^{31} + {}^{32}C_2 9^{30} - {}^{32}C_3 9^{29} + \cdots\cdots - {}^{32}C_{31} 9^1 + {}^{32}C_{32}\right]$

$= 4 \times 9\left[{}^{32}C_0 9^{31} - {}^{32}C_1 9^{30} + {}^{32}C_3 9^{29} - {}^{32}C_4 9^{28} + \ldots\ldots - {}^{32}C_{31}\right] + 4$

$= 9\lambda + 4$

$\therefore$ Remainder is $= 4$.

Q.8 Find the last two digits of the number $(17)^{10}$.

Ans 49

Sol. $(17)^{10} = (289)^5 = (290 - 1)^5$

$= {}^5C_0 (290)^5 - {}^5C_1 (290)^4 + \ldots\ldots + {}^5C_4 (290)^1 - {}^5C_5 (290)^0$

$= {}^5C_0 (290)^5 - {}^5C_1 . (290)^4 + \ldots\ldots {}^5C_3 (290)^2 + 5 \times 290 - 1$

= A multiple of 1000 + 1449

Hence, last two digits are 49

Q.9 Prove that $11^n - 10n - 1$ is always divisible by 100 $\forall$ n $\in$ N.

Ans.

Sol. $\because 11^n - 10n - 1 = \left(1+10\right)^n - 10n - 1$

$= \left[{}^nC_0 + {}^nC_1(10) + {}^nC_2(10)^2 + {}^nC_3(10)^3 + \cdots\cdots + {}^nC_{n-1}(10)^{n-1} + {}^nC_n(10)^n\right] - 10n - 1$

$= 1 + 10n + (10)^2\ {}^nC_2 + (10)^3\ {}^nC_3 + \cdots + (10)^{n-1} \cdot {}^nC_{n-1} + (10)^n \cdot {}^nC_n - 10n - 1$

$= 10^2\left[{}^nC_2 + 10\ {}^nC_3 + (10)^2\ {}^nC_4 + \cdots + 10^{n-2} \cdot {}^nC_n\right]$

$= 100\lambda$

Hence the above expression is always divisible by 100 for every n $\in$ N.

Q.10 Find remainder when $\left(6^{40} + 8^{40}\right)$ is divided by 49.

Ans. 2

Sol. $\because 6^{40} + 8^{40} = (7-1)^{40} + (7+1)^{40}$

$= \left[{}^{40}C_0 7^{40} - {}^{40}C_1 7^{39} + {}^{40}C_2 7^{38} - {}^{40}C_3 7^{37} + \cdots\cdots + {}^{40}C_{38} 7^2 - {}^{40}C_{39} 7^1 + {}^{40}C_{40}\right]$

$+ \left[{}^{40}C_0 7^{40} + {}^{40}C_1 7^{39} + {}^{40}C_2 7^{38} + {}^{40}C_3 7^{37} + \cdots\cdots\cdots + {}^{40}C_{38} 7^2 + {}^{40}C_{39} 7^1 + {}^{40}C_{40}\right]$

$= 2\left[{}^{40}C_0 7^{40} + {}^{40}C_2 7^{38} + {}^{40}C_4 7^{36} + \cdots\cdots\cdots + {}^{40}C_{38} 7^2 + {}^{40}C_{40}\right]$

$= 2 \cdot 7^2\left[{}^{40}C_0 7^{38} + {}^{40}C_2 7^{36} + {}^{40}C_4 7^{34} + \cdots\cdots + {}^{40}C_{38}\right] + 2$

$= 49\lambda + 2$; $\lambda \in z$

$\therefore$ Remainder is = 2.

Q.11 If $(2021)^{3762}$ is divided by 17, then the remainder is _____.

यदि $(2021)^{3762}$ को 17 से विभाजित करने पर शेषफल _____।

Ans. 4

Sol. $(2021)^{3768} = (2023\text{-}z)^{3762}$

$= 17\,k + 2^{3762}$ $\{k \in I\}$

$= 17\,k + 2^2\,(2^4)^{940}$

$= 17\,k + 4\,(17-1)^{940}$

$= 17\,k + 4\,(17\lambda + 1)$ $\{\lambda \in I\}$

$= 17k + 4\,(17\lambda) + 4$

Remainder = 4

Note:

1. $a^n + b^n$ is always divisible by (a + b), if n is odd natural number.
2. $a^n - b^n$ is always divisible by (a – b) $\forall$ n∈N.
3. $a^n - b^n$ is always divisible by (a + b), if n is even natural number.

4.1.1 Fermat's Theorem

if a is coprime to p, then a^{p-1} leaves a remainder of 1 when divided by a prime number p.

Q.1 The remainder when $(2021)^{2022} + (2022)^{2021}$ is divided by 7 is :

Q.2 The remainder when $7^{2022} + 3^{2022}$ is divided by 5 is :

Q.3 The remainder when $(2021)^{2023}$ is divided by 7 is :

Q.4 The remainder when 3^{2022} is divided by 5 is :

Q.5 The remainder on dividing 5^{99} by 11 is _____ .

4.1.2 Euler's Remainder Theorem

if a is coprime to n, then $a^{\phi(n)}$ leaves a remainder of 1 when divided by n . where $\phi(n)$ is euler's totient function

4.2 INTEGRAL PART AND FRACTIONAL PART BASED CONCEPT

Q.1 Prove that integral part of $\left(2+\sqrt{3}\right)^n$ is odd $\forall\, n\in N$.

Sol $\left(2+\sqrt{3}\right)^n = I + f = {}^nC_0 2^n + {}^nC_1 2^{n-1}\sqrt{3} + {}^nC_2 2^{n-2}\left(\sqrt{3}\right)^2 + ...$

$\left(2-\sqrt{3}\right)^n = 0 + f' = {}^nC_0 2^n - {}^nC_1 2^{n-1}\sqrt{3} + {}^nC_2 2^{n-2}\left(\sqrt{3}\right)^2 + ...$

Add $\Rightarrow$ $I + f + f' = 2\,({}^nC_0 2^n + {}^nC_2 2^{n-2}\left(\sqrt{3}\right)^2 + ...)$

$(0 < f < 1;\ 0 < f' < 1 \Rightarrow 0 < f + f' < 2 \Rightarrow f + f' = 1)$

$I + 1 = \text{Even} \Rightarrow \quad I = \text{odd}$

Q.2 Find integral part of $\left(2+\sqrt{3}\right)^5$.

Ans 723

Sol. $\because \left(2+\sqrt{3}\right)^5 = I + f$

$= {}^5C_0 2^5 + {}^5C_1 2^4\left(\sqrt{3}\right) + {}^5C_2 2^3\left(\sqrt{3}\right)^2 + {}^5C_3 2^2\left(\sqrt{3}\right)^3 + {}^5C_4 2^1\left(\sqrt{3}\right)^4 + {}^5C_5\left(\sqrt{3}\right)^5$ .....(1)

$\left(2-\sqrt{3}\right)^5 = 0 + f^1$

$= {}^5C_0 2^5 - {}^5C_1 2^4\left(\sqrt{3}\right) + {}^5C_2 2^3\left(\sqrt{3}\right)^2 - {}^5C_3 2^2\left(\sqrt{3}\right)^3 + {}^5C_4 2^1\left(\sqrt{3}\right)^4 - {}^5C_5\left(\sqrt{3}\right)^5$ .....(2)

(1) + (2) : -

$I + (f + f') = 2\left[{}^5C_0 2^5 + {}^5C_2 2^3 \cdot 3 + {}^5C_4 \cdot 2 \cdot 9\right]$

$= 2[32 + 10 \times 8 \times 3 + 5 \times 2 \times 9]$

$I + (f + f') = 2[32 + 240 + 90] = 2 \times 362 = 724$

$\because\ f + f' = 1$

$\Rightarrow I = 724 - 1 = 723$

Q.3 If $\left(2+\sqrt{3}\right)^{11} = I + f$ then find $(I+f)(1-f)$.

Ans 1

Sol. $\because \left(2+\sqrt{3}\right)^{11} = I + f$ .....(1)

and $\left(2-\sqrt{3}\right)^{11} = f'$ $\quad \because 0 < f < 1$

$0 < f' < 1$

$\Rightarrow 0 < f < f' < 2$

$\Rightarrow f + f' = 1$

$\Rightarrow f' = (1 - f)$

$\Rightarrow \left(2-\sqrt{3}\right)^{11} = (1-f)$ .....(2)

$\therefore \left(2+\sqrt{3}\right)^{11} \cdot \left(2-\sqrt{3}\right)^{11} = (I+f)(1-f)$

$\Rightarrow (I+f)\cdot(1-f') = (4-3)^{11} = (1)^{11} = 1$

Q.4 Prove that integral part of $\left(3\sqrt{3}+5\right)^{2n+1}$ is even $\forall\ n \in N$.

Sol. $\left(3\sqrt{3}+5\right)^{2n+1} = I+f = {}^{2n+1}C_0\left(3\sqrt{3}\right)^{2n+1} + {}^{2n+1}C_1\left(3\sqrt{3}\right)^{2n}\cdot(5) + {}^{2n+1}C_2\left(3\sqrt{3}\right)^{2n-1}(5)^2$

$+{}^{2n+1}c_3\left(3\sqrt{3}\right)^{2n-2}(5)^3 + \cdots\cdots + {}^{2n+1}C_{2n}\left(3\sqrt{3}\right)^{1}(5)^{2n} + {}^{2n+1}C_{2n+1}(5)^{2n+1}$ .....(1)

and $\left(3\sqrt{3}-5\right)^{2n+1} = f' = {}^{2n+1}C_0\left(3\sqrt{3}\right)^{2n+1} - {}^{2n+1}C_1\left(3\sqrt{3}\right)^{2n}.(5) +$

$+{}^{2n+1}C_2\left(3\sqrt{3}\right)^{2n-1}(5)^2 - {}^{2n+1}C_3\left(3\sqrt{3}\right)^{2n-2}(5)^3 + \cdots\cdots + {}^{2n+1}C_{2n}\left(3\sqrt{3}\right)(5)^{2n} - {}^{2n+1}C_{2n+1}(5)^{2n+1}$ .....(2)

(1) – (2)

$I+(f-f') = 2\left[\ {}^{2n+1}C_1(5)\left(3\sqrt{3}\right)^{2n} + {}^{2n+1}C_3(5)^3\left(3\sqrt{3}\right)^{2n-2} + \ldots\ldots + {}^{2n+1}C_{2n+1}(5)^{2n+1}\right]$

$\because -1 < f-f' < 1$ and $I+(f-f') \in Z$

$\Rightarrow f-f' = 0$

$I + 0 = 2 \times$ Integer

$\Rightarrow I =$ Integer

So it's Integral part is even.

5. MULTINOMIAL THEOREM :

Using binomial theorem, we have $(x+a)^n = \sum_{r=0}^{n} {}^nC_r x^{n-r}a^r,\ n \in N$

$= \sum_{r=0}^{n}\frac{n!}{(n-r)!r!}x^{n-r}a^r = \sum_{r+s=n}\frac{n!}{r!s!}x^s a^r$, where $s+r=n$

This result can be generalized in the following form.

$(x_1+x_2+\ldots\ldots+x_k)^n = \sum_{r_1+r_2+\ldots\ldots+r_k=n}\frac{n!}{r_1!r_2!\ldots\ldots r_k!}x_1^{r_1}x_2^{r_2}\ldots\ldots x_k^{r_k}$

The general term in the above expansion $\frac{n!}{r_1!r_2!r_3!\ldots\ldots r_k!}.x_1^{r_1}x_2^{r_2}x_3^{r_3}\ldots\ldots x_k^{r_k}$

The number of terms in the above expansion is equal to the number of non-negative integral solution of the equation

$r_1+r_2+\ldots\ldots+r_k = n$

Q.1 Find coefficient of $x^2y^3z^5$ in the expansion of $(x+y+z)^{10}$.

Ans 2520

Sol. $\because (x+y+z)^{10} = \sum_{r_1+r_2+r_3=10}\frac{10!}{r_1!r_2!r_3!}x^{r_1}\cdot y^{r_2}z^{r_3}$

So for coeff of $x^2 . y^3 . z^5$ put $r_1 = 2, r_2 = 3, r_3 = 5$

$\therefore$ Coeff of $x^2\cdot y^3\cdot z^5 = \frac{10!}{2!3!5!} = 2520$

Q.2 Find the coefficient of $a^2 b^3 c^4 d$ in the expansion of $(a - b - c + d)^{10}$

Ans -12600

Sol. $(a - b - c + d)^{10} = \sum_{r_1+r_2+r_3+r_4=10} \frac{(10)!}{r_1!\, r_2! r_3!\, r_4!} (a)^{r_1} (-b)^{r_2} (-c)^{r_3} (d)^{r_4}$

we want to get $a^2 b^3 c^4 d$ this implies that $r_1 = 2, r_2 = 3, r_3 = 4, r_4 = 1$

$\therefore$ coeff. of $a^2 b^3 c^4 d$ is $\frac{(10)!}{2!\, 3!\, 4!\, 1!} (-1)^3 (-1)^4 = -12600$

Q.3 Find coefficient of x^2 in the expansion of $\left(2 + x - \frac{7}{x}\right)^7$.

Ans. 3122

Sol. $\therefore$ General term $= \frac{7!}{r_1!\, r_2!\, r_3!} (2)^{r_1} (x)^{r_2} \cdot \left(-\frac{7}{x}\right)^{r_3}$

$= \frac{7!}{r_1!\, r_2!\, r_3!} 2^{r_1} (-7)^{r_3} (x)^{r_2 - r_3}$

$\therefore$ for coeff of x^2, put $r_2 - r_3 = 2$

r_1	r_2	r_3
5	2	0
3	3	1
1	4	2

$\therefore$ coeff of $x^2 = \left(\frac{7!}{5!2!0!}\right) 2^5 (-7)^0 + \frac{7!}{3!3!1!} 2^3 (-7)^1 + \frac{7!}{1!4!2!} 2^1 (-7)^2$

$= \frac{7 \times 6}{2} \times 32 + \frac{7 \times 6 \times 5 \times 4}{6} \times 8(-7) + \frac{7 \times 6 \times 5}{1} \times 49$

$= 3122$

Q.4 Find coefficient of x^3 in expansion of $(1 + 3x + 3x^2 + x^3)^{10}$

Ans 4060

Sol. $[(1 + 3x + 3x^2 + x^3)]^{10}$

$= [(1 + x^3) + 3x(1 + x)]^{10}$

$= [(1 + x).(1 - x + x^2) + 3x(1 + x)]^{10}$

$= [(1 + x)(1 - x + x^2 + 3x)]^{10}$

$= [(1 + x)(1 + 2x + x^2)]^{10}$

$= [(1 + x)(1 + x)^2]^{10} = [(1 + x)^3]^{10} = (1 + x)^{30}$

$= (1 + x)^{30}$

$\therefore$ Coeff.of $x^3 = {}^{30}c_3 = \frac{30!}{3!27!} = \frac{30 \times 29 \times 28}{6}$

$= 5 \times 29 \times 28 = 4060$

6.1 SUMMATION OF SERIES TYPE - 1

$$(x+y)^n = \sum_{r=0}^{n} {}^nC_r x^{n-r} y^r$$

1. $\sum_{r=0}^{7} {}^7C_r 2^{7-r} 3^r$

Ans. $(2+3)^7$

2. $\sum_{r=0}^{7} {}^7C_r 2^r$

Ans. $(1+2)^7$

3. $\sum_{r=0}^{7} {}^7C_r 3^{7-r}$

Ans. $(3+1)^7$

4. $\sum_{r=0}^{n} {}^nC_r$

Ans. $(1+1)^n$

5. $\sum_{r=0}^{7} {}^7C_r 3^{2r}$

Ans. $(1+9)^7$

6. $\sum_{r=0}^{n} {}^nC_r 5^{r+1}$

Ans. $5(1+5)^n$

Sol. $= \sum_{r=0}^{n} {}^nC_r \cdot 5^r \cdot 5^1 = 5\sum_{r=0}^{n} {}^nC_r 5^r$

$= 5(1+5)^n = 5 \cdot 6^n$

7. $\sum_{r=1}^{n} {}^nC_r 2^r$

Ans. $3^n - 1$

Sol. $= \left[{}^nC_1 2 + {}^nC_2 \cdot 2^2 + {}^nC_3 2^3 + {}^nC_4 2^4 + \cdots + {}^nC_n 2^n \right] + {}^nC_0 - {}^nC_0$

$= \left[{}^nC_0 + {}^nC_4 (2) + {}^nC_2 \cdot (2)^2 + {}^nC_3 (2)^3 + \cdots + {}^nC_n (2)^n \right] - 1$

$= (1+2)^n - 1 = 3^n - 1$

8. $\sum_{r=1}^{n} {}^nC_r \left(a^r - 1\right)$

Ans. $(1+a)^n - 2^n$

Sol. $= \sum_{r=1}^{n} {}^nC_r (a)^r - \sum_{r=1}^{n} {}^nC_r$

$= (1 + a)^n - 2^n$

6.2 SUMMATION OF SERIES TYPE - 2

Upper suffix varies

${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

1. Prove that : ${}^{25}C_{10} + {}^{24}C_{10} + \ldots\ldots\ldots + {}^{10}C_{10} = {}^{26}C_{11}$

Sol. LHS $= {}^{10}C_{10} + {}^{11}C_{10} + {}^{12}C_{10} + \ldots\ldots\ldots\ldots + {}^{25}C_{10}$

$\Rightarrow \quad = {}^{11}C_{11} + {}^{11}C_{10} + {}^{12}C_{10} + \ldots\ldots + {}^{25}C_{10}$

$\Rightarrow \quad = {}^{12}C_{11} + {}^{12}C_{10} + \ldots\ldots\ldots + {}^{25}C_{10}$

$\Rightarrow \quad = {}^{13}C_{11} + {}^{13}C_{10} + \ldots\ldots\ldots {}^{25}C_{10}$

and so on. $\therefore$ LHS $= {}^{26}C_{11}$

Aliter :

LHS = coefficient of x^{10} in $\{(1 + x)^{10} + (1 + x)^{11} + \ldots\ldots\ldots\ldots (1+ x)^{25}\}$

$\Rightarrow$ coefficient of x^{10} in $\left[(1 + x)^{10} \dfrac{\{1 + x\}^{16} - 1}{1 + x - 1}\right]$

$\Rightarrow$ coefficient of x^{10} in $\dfrac{\left[(1 + x)^{26} - (1 + x)^{10}\right]}{x}$

$\Rightarrow$ coefficient of x^{11} in $\left[(1 + x)^{26} - (1 + x)^{10}\right] = {}^{26}C_{11} - 0 = {}^{26}C_{11}$

2. ${}^nC_n + {}^{n+1}C_n + {}^{n+2}C_n + \ldots\ldots\ldots + {}^{n+r}C_n$

Ans ${}^{n+r+1}C_{n+1}$

Sol. ${}^nC_n + {}^{n+1}C_n + {}^{n+2}C_n + \ldots\ldots\ldots + {}^{n+r}C_n$

$= {}^{n+1}C_{n+1} + {}^{n+2}C_n + {}^{n+3}C_n + \cdots\cdots + {}^{n+r}C_n$

$= {}^{n+2}C_{n+1} + {}^{n+3}C_n + \cdots\cdots + {}^{n+r}C_n$

$\vdots$

$= {}^{n+r+1}C_{n+1}$

3. ${}^nC_0 + {}^{n+1}C_1 + {}^{n+2}C_2 + \ldots\ldots\ldots + {}^{n+r}C_r$

Ans ${}^{n+r+1}C_r$

Sol. $= {}^{n+1}C_0 + {}^{n+1}C_1 + {}^{n+2}C_2 + {}^{n+3}C_3 + \cdots\cdots + {}^{n+r}C_r$

$= {}^{n+2}C_1 + {}^{n+2}C_2 + {}^{n+3}C_3 + \cdots\cdots + {}^{n+r}C_r$

$= {}^{n+3}C_2 + {}^{n+3}C_3 + \cdots\cdots + {}^{n+r}C_r$

$\vdots$

$= {}^{n+r+1}C_r.$

6.3 SUMMATION OF SERIES TYPE - 3

$(1 + x)^n = C_0 + C_1x + C_2x^2 + \ldots\ldots + C_nx^n$ (where, $C_r = {}^nC_r$)

Put $x = 1 \Rightarrow \quad 2^n = C_0 + C_1 + C_2 + \ldots\ldots\ldots + C_n \ldots\ldots$ (i)

Put $x=-1 \Rightarrow \quad 0 = C_0 - C_1 + C_2 + \ldots\ldots\ldots \quad \ldots\ldots\ldots$ (ii)

(i) + (ii) $\Rightarrow \quad \dfrac{2^n+0}{2} = C_0 + C_2 + C_4 + \ldots\ldots\ldots\ldots$ (iii)

(i) – (ii) $\Rightarrow \quad \dfrac{2^n-0}{2} = C_1 + C_3 + C_5 + \ldots\ldots\ldots\ldots$ (iv)

1. To find sum of coefficient of even powers of x, put x = 1, x = –1 and add both the expressions.
2. To find sum of coefficient of odd powers of x, put x = 1, x = –1 and subtract both the expressions.

Q.1 $S = 1\left({}^nC_0\right)+2\left({}^nC_1\right)+3\left({}^nC_2\right)+\ldots\ldots\ldots\ldots\ldots+(n+1)\left({}^nC_n\right)$ or $\sum_{r=0}^{n}(r+1)\,{}^nC_r$

Ans. $(n+2)\cdot 2^{n-1}$

Sol. $S = \sum_{r=0}^{n}(r+1)\cdot {}^nC_r$

$= \sum_{r=0}^{n} r\cdot {}^nC_r + \sum_{r=0}^{n} {}^nC_r$

$= \sum_{r=0}^{n} n\cdot {}^{n-1}C_{r-1} + 2^n$

$= n\sum_{r=0}^{n} {}^{n-1}C_{r-1} + 2^n$

$= n\left[{}^{n-1}C_0 + {}^{n-1}C_1 + {}^{n-1}C_2 + {}^{n-1}C_3 + \cdots + {}^{n-1}C_{n-1}\right] + 2^n$

$= n\cdot 2^{n-1} + 2^n = n\cdot 2^{n-1} + 2\cdot 2^{n-1}$

$= (n+2)\cdot 2^{n-1}.$

Q.2 $S = {}^nC_0 + (4)\left({}^nC_1\right)+7\left({}^nC_2\right) + \ldots\ldots\ldots\ldots\ldots+(3n+1)\left({}^nC_n\right)$ or $\sum_{r=0}^{n}(3r+1)\,{}^nC_3$

Ans. $(3n+2)\,2^{n-1}$

Sol. $S = \sum_{r=0}^{n}(3r+1)\cdot {}^nc_r$

$S = 3\sum_{r=0}^{n} r\cdot {}^nc_r + \sum_{r=0}^{n} {}^nc_r.$

$S = 3\sum_{r=1}^{n} n.\,{}^{n-1}c_{r-1} + 2^n$

$S = 3n\sum_{r=1}^{n} {}^{n-1}C_{r-1} + 2^n$

$S = 3n\cdot 2^{n-1} + 2^n = 2^n\left[\dfrac{3n}{2}+1\right]$

$$S = 2^n \cdot \frac{(3n+2)}{2} = (3n+2)2^{n-1}$$

Q.3 $S = {}^nC_1 + 2\left({}^nC_2\right) + 3\left({}^nC_2\right) + \ldots\ldots\ldots\ldots + n\left({}^nC_n\right)$ or $\sum_{r=0}^{n} r\,{}^nC_r$

Ans. $n2^{n-1}$

Sol $\text{L.H.S.} = \sum_{r=1}^{n} r.\,{}^nC_r = \sum_{r=1}^{n} r.\frac{n}{r}.\,{}^{n-1}C_{r-1}$

$$= n\sum_{r=1}^{n} {}^{n-1}C_{r-1} = n.\left[{}^{n-1}C_0 + {}^{n-1}C_1 + \ldots\ldots + {}^{n-1}C_{n-1}\right]$$

$$= n \,.\, 2^{n-1}$$

Q.4 $\sum_{r=0}^{n} r^2\left({}^nC_r\right)$

Ans. $n(n+1)\,2^{n-2}$

Sol. $\sum_{r=0}^{n} r\cdot\left(r\cdot{}^nC_r\right) = \sum_{r=1}^{n} r\cdot\left(n\cdot{}^{n-1}C_{r-1}\right)$

$$= n\sum_{r=1}^{n} r\cdot{}^{n-1}C_{r-1} = n\sum_{r=1}^{n}(r-1+1)\,{}^{n-1}C_{r-1}$$

$$= n\left[\sum_{r=1}^{n}(r-1)\cdot{}^{n-1}C_{r-1} + \sum_{r=1}^{n}{}^{n-1}C_{r-1}\right]$$

$$= n\left[\sum_{r=2}^{n}(n-1)\cdot{}^{n-2}C_{r-2} + 2^{n-1}\right]$$

$$= n(n-1)\sum_{r=2}^{n}{}^{n-2}C_{r-2} + n\cdot 2^{n-1}$$

$$= n(n+1)\cdot 2^{n-2} + n\cdot 2^{n-1}$$

$$= n\cdot 2^{n-2}[n-1+2] = n(n+1)\cdot 2^{n-2}$$

6.4 SUMMATION OF SERIES TYPE - 4

Q.1 Prove that : $C_1 + 2C_2 + 3C_3 + \ldots\ldots + nC_n = n \,.\, 2^{n-1}$

Sol $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \ldots\ldots + {}^nC_nx^n$ .....(A)

Differentiating (A), we get

$n(1+x)^{n-1} = C_1 + 2C_2x + 3C_3x^2 + \ldots\ldots + n.C_nx^{n-1}$.

Put $x = 1$,

$$C_1 + 2C_2 + 3C_3 + \ldots\ldots + n.C_n = n.2^{n-1}$$

Q.2 Prove that $C_0 + 2C_1 + 3.\,C_2 + \ldots\ldots + (n+1)\,C_n = 2^{n-1}(n+2)$.

Sol $(1+x)^n = C_0 + C_1x + C_2x^2 + \ldots\ldots\ldots + C_nx^n$

Multiplying both sides by x,

$x(1 + x)^n = C_0x + C_1x^2 + C_2x^3 + \ldots\ldots + C_n x^{n+1}$.

Differentiating both sides

$(1 + x)^n + x\, n\, (1 + x)^{n-1} = C_0 + 2.\, C_1x + 3\, .\, C_2x^2 + \ldots\ldots + (n + 1)C_nx^n$.

putting x = 1, we get

$C_0 + 2.C_1 + 3\, .\, C_2 + \ldots\ldots + (n + 1)\, C_n = 2^n + n\, .\, 2^{n-1}$

$C_0 + 2.C_1 + 3\, .\, C_2 + \ldots\ldots + (n + 1)\, C_n = 2^{n-1}\, (n + 2)$

Q.3 Find $1^2.\, C_1 + 2^2.\, C_2 + 3^2.\, C_3 + \ldots + n^2.\, C_n$

Ans. $n(n + 1)2^{n-2}$

Sol. We have

$(1 + x)^n = C_0 + C_1x + C_2x^2 + \ldots + C_nx^n$ . . . (i)

Differentiating equation (i) w.r.t. x, we have

$n(1 + x)^{n-1} = C_1 + 2C_2x + 3C_3x^2 + \ldots + nC_nx^{n-1}$ . . . (ii)

Multiplying equation (ii) throughout by x, we have

$nx(1 + x)^{n-1} = C_1x + 2C_2x^2 + 3C_3x^3 + \ldots + nC_nx^n$ . . . (iii)

Differentiating equation (iii) w.r.t. x, we have

$n(1 + x)^{n-1} + n\,(n - 1)(1 + x)^{n-2}$

$= C_1 + 2^2.\, C_2x + 3^2.\, C_3x^2 + \ldots + n^2.\, C_n\, x^{n-1}$ . . . (iv)

Putting x = 1 in equation (iii), we have

$1^2.\, C_1 + 2^2.\, C_2 + 3^2.\, C_3 + \ldots + n^2.\, C_n$

$= n.\, 2^{n-1} + n(n - 1).\, 2^{n-2} = (n^2 + n)\, 2^{n-2}$

$= n\,(n + 1)2^{n-2}$.

Q.4 Prove that $C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \ldots\ldots\ldots + \frac{C_n}{n+1} = \frac{2^{n+1} - 1}{n+1}$

Sol. $\because (1+x)^n = {}^nC_0 + {}^nC_1 \cdot x + {}^nC_2 \cdot x^2 + {}^nC_3x^3 + \cdots\cdots + {}^nC_nx^n$

Integrate both sides with respect tox, from 0 to 1 : -

$$\int_0^1 (1+x)^n \cdot dx = {}^nC_0\int_0^1 1 \cdot dx + {}^nC_1\int_0^1 x \cdot dx + {}^nC_2\int_0^1 x^2 \cdot dx + {}^nC_3\int_0^1 x^3 \cdot dx$$

$$+ \cdots\cdots + {}^nC_n\int_0^1 x^n \cdot dx$$

$$\left[\frac{(1+x)^{n+1}}{n+1}\right]_0^1 = {}^nC_0[x]_0^1 + {}^nC_1\left[x^2/2\right]_0^1 + {}^nC_2\left[\frac{x^3}{3}\right]_0^1 + {}^nC_3\left[x^4/4\right]_0^1 + \cdots\cdots + {}^nC_n\left[\frac{x^{n+1}}{n+1}\right]_0^1$$

$$\frac{2^{n+1}}{n+1} - \frac{1}{n+1} = {}^nC_0(1) + {}^nC_1\left(\frac{1}{2}\right) + {}^nC_2\left(\frac{1}{3}\right) + \cdots + {}^nC_n\left(\frac{1}{n+1}\right)$$

$$\therefore {}^{n}C_0 + \frac{{}^{n}C_1}{2} + \frac{{}^{n}C_2}{3} + \cdots + \frac{{}^{n}C_n}{n+1} = \frac{1}{n+1}\left(2^{n+1} - 1\right) = R \cdot H \cdot S.$$

Q.5 $${}^{n}C_0(2) + \frac{{}^{n}C_1(2)^2}{2} + \frac{{}^{n}C_2(2)^3}{3} + \ldots\ldots\ldots\ldots\ldots + \frac{{}^{n}C_n 2^{n+1}}{n+1}.$$

Ans. $$\frac{3^{n+1} - 1}{n+1}$$

Sol. $$\because (1+x)^n = {}^{n}C_0 + {}^{n}C_1 x + {}^{n}C_2 x^2 + {}^{n}C_3 x^3 + \cdots + {}^{n}C_n \cdot x^n$$

Integrate both sides with respect to 'x' from 0 to 2 : -

$$\int_0^2 (1+x)^n \cdot dx = {}^{n}C_0 \int_0^2 dx + {}^{n}C_1 \int_0^2 x \cdot dx + {}^{n}C_2 \int_0^2 x^2 \cdot dx$$

$$+ {}^{n}C_3 \int_0^2 x^3 \cdot dx + \cdots\cdots\cdots + {}^{n}C_n \int_0^2 x^n \cdot dx$$

$$\left[\frac{(1+x)^{n+1}}{n+1}\right]_0^2 = {}^{n}C_0 [x]_0^2 + {}^{n}C_1 \left[\frac{x^2}{2}\right]_0^2 + {}^{n}C_2 \left[\frac{x^3}{3}\right]_0^2 + \cdots\cdots + {}^{n}C_n \left[\frac{x^{n+1}}{n+1}\right]_0^2$$

$$\frac{1}{n+1}\left[3^{n+1} - 1\right] = {}^{n}C_0(2-0) + {}^{n}C_1\left(\frac{2^2}{2}\right) + {}^{n}C_2\left(\frac{2^3}{3}\right) + \cdots + {}^{n}C_n \frac{2^{n+1}}{n+1}.$$

$$\therefore {}^{n}C_0(2) + {}^{n}C_1 \cdot \frac{\left(2^2\right)}{2} + {}^{n}C_2 \cdot \frac{2^3}{3} + {}^{n}C_3 \cdot \frac{2^4}{4} + \cdots + {}^{n}C_n \cdot \frac{2^{n+1}}{n+1}$$

$$= \frac{1}{n+1}\left[3^{n+1} - 1\right]$$

Q.6 $$\sum_{r=0}^{n} \frac{{}^{n}C_r}{r+1}$$

Ans. $$\frac{2^{n+1} - 1}{n+1}$$

Sol. $$\sum_{r=0}^{n} \frac{{}^{n}C_r}{r+1} = \sum_{r=0}^{n} \frac{1}{n+1} \cdot {}^{n+1}C_{r+1} \quad \left\{\because {}^{n+1}C_{r+1} = \frac{n+1}{r+1} \cdot {}^{n}C_r\right\}$$

$$= \frac{1}{n+1} \sum_{r=0}^{n} {}^{n+1}C_{r+1}$$

$$= \frac{1}{n+1}\left[{}^{n+1}C_1 + {}^{n+1}C_2 + {}^{n+1}C_3 + \cdots + {}^{n+1}C_{n+1}\right]$$

$$= \frac{1}{n+1}\left[{}^{n+1}C_0 + {}^{n+1}C_1 + {}^{n+1}C_2 + {}^{n+1}C_3 + \cdots + {}^{n+1}C_{n+1} - {}^{n+1}c_0\right]$$

$$= \frac{1}{n+1}\left[2^{n+1} - 1\right]$$

6.5 SUMMATION OF SERIES TYPE - 5

${}^{n}C_{r_1}\ {}^{m}C_{r_2}$ where $r_1 + r_2 =$ constant , $r_1 - r_2 =$ constant.

1. ${}^{10}C_0\,{}^{15}C_{12} + {}^{10}C_1\,{}^{15}C_{11} + {}^{10}C_2\,{}^{15}C_{10} + \ldots\ldots\ldots\ldots\ldots + {}^{10}C_{10}\,{}^{15}C_2$.

Ans. ${}^{25}C_{12}$

Sol. $\because (1+x)^{10} = {}^{10}C_0 + {}^{10}C_1x^1 + {}^{10}C_2x^2 + {}^{10}C_3x^3 + \cdots\cdots\cdots + {}^{10}C_{10}x^{10}$

and $(1+x)^{15} = {}^{15}C_0 + {}^{15}C_1x + {}^{15}C_2x^2 + {}^{15}C_3x^3 + \cdots\cdots + {}^{15}C_{15}x^{15}$

Now, multiply both equation and compare coeff of x^{12} both sides, we get :-

${}^{10}C_0 \cdot {}^{15}C_{12} + {}^{10}C_1 \cdot {}^{15}C_{11} + {}^{10}C_2 \cdot {}^{15}C_{10} + \cdots\cdots + {}^{10}C_{10}\,{}^{15}C_2$

= coeff. of x^{12} in the exp. of $(1+x)^{10+15}$

$= {}^{25}C_{12}$

2. ${}^{n}C_0\,{}^{n}C_n + {}^{n}C_1\,{}^{n}C_{n-1} + \ldots\ldots\ldots\ldots + {}^{n}C_n\,{}^{n}C_0$.

Ans. ${}^{2n}C_n$

Sol. $\because (1+x)^n = {}^{n}C_0 + {}^{n}C_1x + {}^{n}C_2x^2 + {}^{n}C_3x^3 + \cdots + {}^{n}C_n \cdot x^n$ .....(1)

$(x+1)^n = {}^{n}C_nx^n + {}^{n}C_{n-1}x^{n-1} + {}^{n}C_{n-2}x^{n-2} + {}^{n}C_{n-3}x^{n-3} + \cdots + {}^{n}C_0$ .....(2)

Now multiply (1) and (2), and compare coeff. of x^n, both sides : -

${}^{n}C_0 \cdot {}^{n}C_n + {}^{n}C_1 \cdot {}^{n}C_{n-1} + {}^{n}C_2 \cdot {}^{n}C_{n-2} + \cdots\cdots + {}^{n}C_n \cdot {}^{n}C_0 = {}^{2n}C_n$

3. ${}^{40}C_0\,{}^{30}C_{20} + {}^{40}C_1\,{}^{30}C_{19} + \ldots\ldots\ldots\ldots\ldots + {}^{40}C_{20}\,{}^{30}C_0$.

Ans. ${}^{70}C_{20}$

Sol. $\because (1+x)^{40} = {}^{40}C_0 + {}^{40}C_1x + {}^{40}C_2x^2 + {}^{40}C_3x^3 + \cdots\cdots + {}^{40}C_{39}x^{39} + {}^{40}C_{40}x^{40}$ .....(1)

and $(1+x)^{30} = {}^{30}C_0 + {}^{30}C_1x + {}^{30}C_2x^2 + {}^{30}C_3x^3 +$

$\cdots\cdots + {}^{30}C_{19}x^{19} + {}^{30}C_{20}x^{20} + \cdots\cdots + {}^{30}C_{30}x^{30}$ .....(2)

multiply (1) and (2), and compare coeff. of x^{20}, both sides : -

coeff. of x^{20} in $(1+x)^{70}$ = Coeff. of x^{20} in

$\left[\left({}^{40}C_0 + {}^{40}C_1x + \cdots + {}^{40}C_{40}x^{40}\right)\cdot\left({}^{30}C_0 + {}^{30}C_1x + \cdots + {}^{30}C_{30}x^{30}\right)\right]$

${}^{40}C_0 \cdot {}^{30}C_{20} + {}^{40}C_1 \cdot {}^{30}C_{19} + {}^{40}C_2 \cdot {}^{30}C_{18} + \cdots\cdots + {}^{40}C_{20} \cdot {}^{30}C_0 = {}^{70}C_{20}$

4. ${}^{20}C_0\,{}^{30}C_1 + {}^{20}C_1\,{}^{30}C_2 + \ldots\ldots\ldots\ldots\ldots + {}^{20}C_{20}\,{}^{30}C_{21}$.

Ans. ${}^{50}C_{29}$

Sol. $\because (1+x)^{20} = {}^{20}C_0 + {}^{20}C_1x + {}^{20}C_2x^2 + {}^{20}C_3x^3 + \cdots\cdots + {}^{20}C_{20}x^{20}$ .....(1)

and $(x+1)^{30} = {}^{30}C_0x^{30} + {}^{30}C_1x^{29} + {}^{30}C_2x^{28} + {}^{30}C_3x^{27} + \cdots\cdots + {}^{30}C_{29}x + {}^{30}C_{30}$. .....(2)

Multiply both equation and compare coeff. of x^{29} both sides we get :-

Coeff. of x^{29} in $(1+x)^{50} = {}^{20}C_0 \cdot {}^{30}C_1 + {}^{20}C_1 \cdot {}^{30}C_2 + \cdots + {}^{20}C_{20} \cdot {}^{30}C_{21}$

$\therefore\ {}^{20}C_0 \cdot {}^{30}C_1 + {}^{20}C_1 \cdot {}^{30}C_2 + {}^{20}C_2 \cdot {}^{30}C_3 + \cdots\cdots + {}^{20}C_{20} \cdot {}^{30}C_{21} = {}^{50}C_{29}$

5. ${}^{n}C_0\,{}^{n}C_0 + {}^{n}C_1\,{}^{n}C_1 + {}^{n}C_2\,{}^{n}C_2 + \ldots\ldots\ldots\ldots + {}^{n}C_n\,{}^{n}C_n$.

Ans. $^{2n}C_n$

Sol $(1 + x)^n = C_0 + C_1x + C_2x^2 + \ldots\ldots\ldots + C_n x^n$. .....(i)

$(x + 1)^n = C_0x^n + C_1x^{n-1} + C_2x^{n-2} + \ldots\ldots + C_n x^0$ .....(ii)

Multiplying (i) and (ii)

$(C_0 + C_1x + C_2x^2 + \ldots\ldots\ldots + C_nx^n)\,(C_0x^n + C_1x^{n-1} + \ldots\ldots\ldots + C_nx^0) = (1 + x)^{2n}$

Comparing coefficient of x^n,

$C_0^2 + C_1^2 + C_2^2 + \ldots\ldots\ldots + C_n^2 = {}^{2n}C_n$

6. $^{20}C_0\,{}^{25}C_2 + {}^{20}C_1\,{}^{25}C_3 + \ldots\ldots\ldots\ldots\ldots + {}^{20}C_{20}\,{}^{25}C_{22}$.

Ans. $^{45}C_{22}$

Sol. $\because (1+x)^{20} = {}^{20}C_0 + {}^{20}C_1x + {}^{20}C_2 \cdot x^2 + {}^{20}C_3x^3 + \cdots\cdots + {}^{20}C_{20}x^{20}$ .....(1)

and $(x+1)^{25} = {}^{25}C_0x^{25} + {}^{25}C_1x^{24} + {}^{25}C_2x^{23} + \cdots\cdots + {}^{25}C_{25}x^0$ .....(2)

Multiply both equation and compare coeff. of x^{23} both sides we get :-

Coeff. of x^{23} in $(1 + x)^{45} = {}^{20}C_0 \cdot {}^{25}C_2 + {}^{20}C_1 \cdot {}^{25}C_3 + \cdots\cdots {}^{20}C_2 \cdot {}^{25}C_{22}$

$\Rightarrow {}^{20}C_0 \cdot {}^{25}C_2 + {}^{20}C_1 \cdot {}^{25}C_3 + \cdots\cdots + {}^{20}C_{20} \cdot {}^{25}C_{22} = {}^{45}C_{22}$

6.6 MISCELLANEOUS QUESTIONS ON SUMMATION OF SERIES

1. $\left(1 + x + x^2\right)^n = a_0 + a_1x + a_2x^2 + \ldots\ldots\ldots + a_{2n}x^{2n}$.

(i) $a_0 + a_1 + a_2 + \ldots\ldots\ldots + a_{2n} = ?$ (ii) $a_0 - a_1 + a_2 - a_3 + \ldots\ldots\ldots\ldots + a_{2n} = ?$

(iii) $a_0 + a_2 + a_4 + \ldots\ldots\ldots\ldots + a_{2n} = ?$ (iv) $a_1 + a_3 + a_5 + \ldots\ldots\ldots\ldots a_{2n-1} = ?$

(v) Find coefficient of x^6 in expansion of $\left(1 + x^2 + x^4\right)^n$.

(vi) Find coefficient of x^{11} in expansion of $\left(1 + x^2 + x^4\right)^n$.

(vii) Prove that $a_r = a_{2n-r}\ \forall\ 0 \le r \le n$.

(viii) Prove that $a_0^2 - a_1^2 + a_2^2 - a_3^2 + \ldots\ldots\ldots\ldots + a_{2n}^2 = a_n$.

(ix) Prove that $a_0a_1 - a_1a_2 + a_2a_3 - a_3a_4 + \ldots\ldots\ldots\ldots = 0$

Ans. (i) 3^n (ii) 1 (iii) $\dfrac{3^n + 1}{2}$ (iv) $\dfrac{3^n - 1}{2}$ (v) a_3 (vi) 0

Sol. (i) Put x = 1 in the given expression :-

$(1 + 1 + 1)^n = a_0 + a_1 + a_2 + a_3 + \ldots\ldots\ldots + a_{2n}$

$\therefore\ a_0 + a_1 + a_2 + a_3 + a_4 + \ldots\ldots\ldots + a_{2n} = 3^n$.

(ii) Put x = –1 in given expression :

$\therefore\ a_0 - a_1 + a_2 - a_3 + a_4 - a_5 + \ldots\ldots\ldots + a_{2n} = (1 - 1 + 1)^n = 1$

(iii) Add result (1) and (2), we get :-

$2[a_0 + a_2 + a_4 + a_6 + \ldots\ldots\ldots + a_{2n}] = 3^n + 1$

$\therefore a_0 + a_2 + a_4 + a_6 + \cdots\cdots + a_{2n} = \dfrac{3^n + 1}{2}$

(iv) Subtract repults (1) and (2) :-

$2[a_1 + a_3 + a_5 + a_7 + \ldots\ldots\ldots + a_{2n-1}] = 3^n - 1$

$\therefore a_1 + a_3 + a_5 + a_7 + \cdots + a_{2n-1} = \dfrac{3^n - 1}{2}$

(v) $\because \left(1 + x + x^2\right)^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots\cdots + a_{2n} \cdot x^{2n}$

Replace x by x^2, we get :-

$\left(1 + x^2 + x^4\right)^n = a_0 + a_1 x^2 + a_2 x^4 + a_3 x^6 + \cdots\cdots + a_{2n} x^{4n}$.

Clearly coeff of $x^6 = a_3$.

(vi) $\because \left(1 + x^2 + x^4\right)^n = a_0 + a_1 x^2 + a_2 x^4 + a_3 x^6 + \cdots\cdots + a_{2n} x^{4n}$

Clearly there is only even degree term in the above exprassion.
Hence Coeff. of $x^{11} = 0$

(vii) $\because \left(1 + x + x^2\right)^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots\cdots + a_{2n} x^{2n}$ .....(1)

Replace x by $\dfrac{1}{x}$: -

$$\left(1 + \frac{1}{x} + \frac{1}{x^2}\right)^n = a_0 + a_1\left(\frac{1}{x}\right) + a_2\left(\frac{1}{x}\right)^2 + a_3\left(\frac{1}{x}\right)^3 + \cdots + a_{2n}\left(\frac{1}{x}\right)^{2n}$$

$$\frac{\left(x^2 + x + 1\right)^n}{x^{2n}} = \frac{a_0 x^{2n} + a_1 x^{2n-1} + a_2 x^{2n-2} + a_3 x^{2n-3} + \cdots + a_{2n}}{x^{2n}}$$

$\left(1 + x + x^2\right)^n = a_0 x^{2n} + a_1 x^{2n-1} + a_2 x^{2n-2} + \cdots + a_{2n-2} x^2 + a_{2n-1} x^1 + a_{2n}$ .....(2)

$\therefore$ from (1) and (2) :

$a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots\cdots + a_{2n} x^{2n}$

$= a_{2n} + a_{2n-1} x + a_{2n-2} \cdot x^2 + \cdots\cdots + a_2 \cdot x^{2n-2} + a_1 x^{2n-1} + a_0 x^{2n}$

By comparing coeff. both sides we get :

$a_{2n} = a_0$
$a_{2n-1} = a_1$
$a_{2n-2} = a_2$
$a_{2n-3} = a_3$
$\vdots$
$a_{2n-r} = a_r$

$\therefore a_{2n-r} = a_r$; $0 \le r \le 2n$ H.P.

(viii) $\because \left(1 + x + x^2\right)^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots\cdots + a_{2n} x^{2n}$ .....(1)

Replace x by $-\dfrac{1}{x}$ we get : -

$$\left(1 - \frac{1}{x} + \frac{1}{x^2}\right)^n = a_0 - \frac{a_1}{x} + \frac{a_2}{x^2} - \frac{a_3}{x^3} + \cdots\cdots + \frac{a_{2n}}{x^{2n}}$$

$$\frac{\left(x^2-x+1\right)^n}{x^{2n}}=\frac{a_0x^{2n}-a_1x^{2n-1}+a_2x^{2n-2}-a_3x^{2n-3}+\cdots+a_{2n}}{x^{2n}}$$

$$\left(1-x+x^2\right)^n=a_0x^{2n}-a_1x^{2n-1}+a_2x^{2n-2}-a_3x^{2n-3}+\cdots+a_{2n} \quad \text{.....(2)}$$

Now multiply (1) and (2), and compare coeff of x^{2n} we get :

$$\text{Coeff. fo } x^{2n} \text{ in } \left(\left(1+x+x^2\right)^n\cdot\left(1-x+x^2\right)^n\right)=a_0^2-a_1^2+a_2^2-a_3^2+a_4^2-\cdots\cdots+a_{2n}{}^2$$

$$\begin{aligned} a_0{}^2-a_1{}^2+a_2{}^2-a_3{}^2+a_4{}^2-\cdots+a_{2n}{}^2 &= \text{Coeff. of } x^{2n} \text{ in } [(1+x^2)^2-x^2]^n \\ &= \text{Coeff. of } x^{2n} \text{ in } [1+x^4+2x^2-x^2]^n \\ &= \text{Coeff. of } x^{2n} \text{ in } (1+x^2+x^4)^n \quad \text{.....(3)} \end{aligned}$$

$$\because \left(1+x+x^2\right)^n=a_0+a_1x+a_2x^2+\cdots+a_nx^n+\cdots+a_{2n}x^{2n}$$

replace x by x^2 :

$$\left(1+x^2+x^4\right)^n=a_0+a_1x^2+a_2x^4+\cdots\cdots+a_nx^{2n}+\cdots+a_{2n}x^{4n} \quad \text{.....(4)}$$

$\therefore$ from (4) :

coeff. of x^{2n} in $(1+x^2+x^4)^n = a_n$.

Hence from (3) :

$$a_0^2-a_1^2+a_2^2-a_3^2+a_4^2-a_5^2+\cdots\cdots+a_{2n}^2=a_n.$$

(ix) $\because \left(1+x+x^2\right)^n=a_0+a_1x+a_2x^2+a_3x^3+\cdots\cdots+a_{2n}x^{2n}$ .....(1)

Replace x by $-1/x$:

$$\left(1-\frac{1}{x}+\frac{1}{x^2}\right)^n=a_0-\frac{a_1}{x}+\frac{a_2}{x^2}-\frac{a_3}{x^3}+\frac{a_4}{x^4}-\cdots+\frac{a_{2n}}{x^{2n}}$$

$$\left(x^2-x+1\right)^n=a_0x^{2n}-a_1x^{2n-1}+a_2x^{2n-2}-a_3x^{2n-3}+\cdots+a_{2n} \quad \text{.....(2)}$$

Now multiply (1) and (2), and then compare coeff of x^{2n-1} both sides we get :

$$-a_0a_1+a_1a_2-a_2a_3+a_3a_4-a_1a_5+\cdots\cdots$$

$$\begin{aligned} &= \text{Coeff. of } x^{2n-1} \text{ in } \left[\left(1+x+x^2\right)^n\cdot\left(1-x+x^2\right)^n\right] \\ &= \text{Coeff. of } x^{2n-1} \text{ in } [(1+x^2)^2-x^2]^n \\ &= \text{Coeff. of } x^{2n-1} \text{ in } [1+x^4+2x^2-x^2]^n \\ &= \text{Coeff. of } x^{2n-1} \text{ in } (1+x^2+x^4)^n \end{aligned}$$

$\because$ We know that in the exparsion of $(1+x^2+x^4)^n$

there is no odd degree term.

$$\therefore \text{Coeff.of } x^{2n-1} \text{in} \left(1+x^2+x^4\right)^n=0$$

Hence $-a_0a_1+a_1a_2-a_2a_3+a_3a_4-a_4a_5+\cdots=0$

$$=a_0a_1-a_1a_2+a_2a_3-a_3a_4+a_4a_5-\cdots=0$$

2. If $\alpha_n=\sum_{r=0}^{n}\frac{1}{{}^nC_r}$ find $\sum_{r=0}^{n}\frac{r}{{}^nC_r}$.

Ans. $\frac{n\alpha_n}{2}$

Sol. $\because \alpha_n = \sum_{r=0}^{n} \frac{1}{n_{c_r}}$

Now $S = \sum_{r=0}^{n} \frac{r}{{}^nC_r} = \sum_{r=0}^{n} \frac{n-(n-r)}{{}^nC_r}$

$S = \sum_{r=0}^{n} \frac{n}{{}^nC_r} - \sum_{r=0}^{n} \frac{(n-r)}{{}^nC_r}$

$S = n\sum_{r=0}^{n} \frac{1}{{}^nC_r} - \sum_{r=0}^{n} \frac{(n-r)}{{}^nC_r}$

$S = n\alpha_n - S \Rightarrow 2S = n\alpha_n \Rightarrow S = \frac{n \cdot \alpha_n}{2}$

3. Find coefficient of x^{20} in expansion of $y = 1 + (1+x) + (1+x)^2 + \ldots\ldots\ldots\ldots + (1+x)^{40}$.

Ans. ${}^{41}C_{21}$

Sol. $y = 1 + (1+x) + (1+x)^2 + (1+x)^3 + \ldots\ldots\ldots\ldots + (1+x)^{40}$

$y = 1 \cdot \left[\frac{(1+x)^{41}-1}{1+x-1}\right] = \frac{1}{x}\left[(1+x)^{41}-1\right]$

$y = \frac{1}{x}\left[{}^{41}C_0 + {}^{41}C_1 x + {}^{41}C_2 \cdot x^2 + \cdots\cdots + {}^{41}C_{20}x^{20} + {}^{41}C_{21}x^{21} + \cdots\cdots + {}^{41}C_{41}x^{41} - 1\right]$

$y = \left[{}^{41}C_1 + {}^{41}C_2 x + {}^{41}C_3 x^2 + \cdots\cdots + {}^{41}C_{20}x^{19} + {}^{41}C_{21}x^{20} + \cdots\cdots + {}^{41}C_{41}x^{40}\right]$

$\therefore$ Coeff. of $x^{20} = {}^{41}C_{21}$

4. $\sum_{r=0}^{n} r\left({}^nC_r\right)^2$.

Ans $n\left({}^{2n-1}C_{n-1}\right)$

Sol. Let $S = \sum_{r=0}^{n} r \cdot \left({}^nC_r\right)^2$

$S = \sum_{r=0}^{n} \left(r \cdot {}^nC_r\right) \cdot {}^nC_r$

$S = \sum_{r=0}^{n} n \cdot {}^{n-1}C_{r-1} \cdot {}^nC_r$

$$S = n\sum_{r=0}^{n} {}^{n-1}C_{r-1} \cdot {}^{n}C_{r}$$

$$S = n\left[{}^{n-1}C_0 \cdot {}^{n}C_1 + {}^{n-1}C_1 \cdot {}^{n}C_2 + {}^{n-1}C_2 \cdot {}^{n}C_3 + \cdots + {}^{n-1}C_{n-1} \cdot {}^{n}C_n \right] \quad \text{.....(1)}$$

$$\because (1+x)^n = {}^{n}C_0 + {}^{n}C_1 x + {}^{n}C_2 .x^2 + {}^{n}C_3 .x^3 + \cdots\cdots + {}^{n}C_n .x^n \quad \text{.....(2)}$$

and $(x+1)^{n-1} = {}^{n-1}C_0 x^{n-1} + {}^{n-1}C_1 x^{n-2} + {}^{n-1}C_2 x^{n-3} + \cdots\cdots + {}^{n-1}C_{n-2} x + {}^{n-1}C_{n-1} x^0$ .....(3)

Multiply (2) and (3) and compare coeff of x^n both sides we get.

coeff of $x^n \left[(1+x)^n \cdot (1+x)^{n-1} \right]$

$$= {}^{n-1}C_0 \cdot {}^{n}C_1 + {}^{n-1}C_1 \cdot {}^{n}C_2 + {}^{n-1}C_2 \cdot {}^{n}C_3 + \cdots\cdots + {}^{n-1}C_{n-1} \cdot {}^{n}C_n$$

$$\therefore {}^{n-1}C_0 \cdot {}^{n}C_1 + {}^{n-1}C_2 \cdot {}^{n}C_2 + {}^{n-1}C_2 \cdot {}^{n}C_3 + \cdots\cdots + {}^{n-1}C_{n-1} \cdot {}^{n}C_n = {}^{2n-1}C_n$$

$\therefore$ from result (1) :

$$s = n \cdot {}^{2n-1}C_n = n \cdot {}^{2n-1}C_{n-1}$$

5. Prove that $\prod_{r=1}^{n} \left(\frac{{}^{n}C_r + {}^{n}C_{r-1}}{{}^{n}C_{r-1}} \right) = \frac{(n+1)^n}{n!}$.

Sol. $\because \prod_{r=1}^{n} \frac{{}^{n}C_r + {}^{n}C_{r-1}}{{}^{n}C_{r-1}} = \prod_{r=1}^{n} \frac{{}^{n+1}C_r}{{}^{n}C_{r-1}}$

$$= \prod_{r=1}^{n} \frac{(n+1)!}{r!(n-r+1)!} \times \frac{(r-1)!(n-r+1)!}{n!}$$

$$= \prod_{r=1}^{n} \frac{(n+1)n!(r-1)!}{r(r-1)!n!} = \prod_{r=1}^{n} \frac{n+1}{r}$$

$$= \left(\frac{n+1}{1}\right) \cdot \left(\frac{n+1}{2}\right) \cdot \left(\frac{n+1}{3}\right) \cdot \left(\frac{n+1}{4}\right) \cdot \cdots\cdots \left(\frac{n+1}{n}\right)$$

$$= \frac{(n+1)^n}{1 \cdot 2 \cdot 3 \cdot 4 \cdots\cdots n} = \frac{(n+1)^n}{n!}$$

6. If $\sum_{r=0}^{2n} a_r (x-2)^r = \sum_{r=0}^{2n} b_r (x-3)^r$; $a_k = 1 \ \forall \ k \geq n$ prove that $b_n = {}^{2n+1}C_{n+1}$.

Sol.

7.1 BINOMIAL THEOREM FOR ANY INDEX

If $n \in Q, |x| < 1$

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots\ldots \infty$$

$$(1-x)^n = 1 - nx + \frac{n(n-1)x^2}{|2} - \frac{n(n-1)(n-2)}{|3}x^3 + \ldots\ldots$$

$$(1+x)^{-n} = 1 - nx + \frac{n(n+1)x^2}{|2} - \frac{n(n+1)(n+2)}{|3}x^3 + \ldots\ldots$$

$$(1-x)^{-n} = 1 + nx + \frac{n(n+1)x^2}{\lfloor 2} + \frac{n(n+1)(n+2)}{\lfloor 3} x^3 + ...$$

Note: The coefficient of x^r in $(1-x)^{-n}$ is ${}^{n+r-1}C_r$, $n \in N$

Q.1 In expansion of $(3+2x)^{-3}$ find

(i) range of x if its expansion in increasing powers of x is valid. (ii) coefficient of x^7.

Ans. (i) $x \in \left(\frac{-3}{2}, \frac{3}{2}\right)$ (ii) coefficient $= -\frac{3.4.5.....9}{7!}\left(\frac{2}{3}\right)^7 (3)^{-3}$

Sol. (i) $(3+2x)^{-3} = 3^{-3}\left(1+\frac{2x}{3}\right)^{-3}$

Clearly expansion of abvoe expansion is valid only when

$$\left|\frac{2x}{3}\right| < 1 \quad \Rightarrow \quad |x| < \frac{3}{2}$$

$$\Rightarrow x \in \left(-\frac{3}{2}, \frac{3}{2}\right)$$

(ii) Coeff.of $x^7 = -\frac{3\cdot4\cdot5\cdot6\cdot7\cdot8\cdot9}{7!}\left(\frac{2}{3}\right)^7 \cdot (3)^{-3}$

Q.2 Find coefficient of x^3 in expansion of $\left(1+2x-\frac{3}{x}\right)(1-x)^{-4}$

Ans. –65

Sol. Coeff. of x^3 = $1 \times$ coeff. of x^3
+ $2 \times$ coeff. of x^2
+ $(-3) \times$ coeff. of x^4

$$= 1 \times {}^{4+3-1}C_3 + 2 \times {}^{4+2-1}C_2 + (-3) \times {}^{4+4-1}C_4$$

$$= {}^6C_3 + 2 \times {}^5C_2 - 3 \cdot {}^7C_4$$

$$= \frac{6!}{3!3!} + 2 \times \frac{5!}{2!3!} - 3 \times \frac{7!}{4!3!}$$

$$= \frac{6\times5\times4}{6} + 2 \times \frac{5\times4}{2} - 3 \times \frac{7\times6\times5}{6}$$

$$= 20 + 20 - 105 = -65$$

Q.3 Find coefficient of x^{13} in expansion of $\left(1+x+x^2+.....+x^{11}\right)^3 \left(1+x+x^2+.....+x^{21}\right)^2$

Ans. 2365

Sol. $\left(1+x+x^2+\cdots\cdots+x^{''}\right)^3 \cdot \left(1+x+x^2+x^3+\cdots+x^{21}\right)^2$

$$=\left(\frac{1-x^{12}}{1-x}\right)^3\cdot\left(\frac{1-x^{22}}{1-x}\right)^2$$

$$=\left(1-x^{12}\right)^3\cdot\left(1-x^{22}\right)^2\cdot(1-x)^{-5}$$

$$\downarrow \qquad \downarrow \qquad \downarrow$$

$x^0 \qquad x^0 \qquad x^{13}$

and $x^{12} \qquad x^0 \qquad x^1$

$\therefore$ Coeff. of $x^{13} = 1\times1\times{}^{5+13-1}C_{13}+(-3)\times(1)\times{}^{5+1-1}C_1$

$={}^{17}C_{13}-3\times{}^5C_1$

$=\frac{17!}{13!4!}-3\times5=\frac{17\times16\times15\times14}{4\times3\times2}-15$

$= 17 \times 2 \times 5 \times 14 - 15 = 2365$

Q.4 The sum of $1+\frac{1}{4}+\frac{1.3}{4.8}+\frac{1.3.5}{4.8.12}+....\infty$ is -

(A) $\sqrt{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\sqrt{3}$ (D) $2^{3/2}$

Ans A

Sol. Comparing with $1+nx+\frac{n(n-1)}{2!}x^2+....$

$nx = 1/4$ ..... (i)

and $\frac{n(n-1)x^2}{2!}=\frac{1.3}{4.8}$

or $\frac{nx(nx-x)}{2!}=\frac{3}{32}\Rightarrow\frac{1}{4}\left(\frac{1}{4}-x\right)=\frac{3}{16}$ (by (i))

$\Rightarrow\left(\frac{1}{4}-x\right)=\frac{3}{4}\Rightarrow x=\frac{1}{4}-\frac{3}{4}=-\frac{1}{2}$ .....(ii)

putting the value of x in (i)

$n(-1/2) = 1/4 \Rightarrow n = -1/2$

$\therefore$ sum of series $= (1 + x)^n = (1 - 1/2)^{-1/2} = (1/2)^{-1/2} = \sqrt{2}$

7.2 OTHER SERIES

1. Exponential series

(a) e is an irrational number lying between 2.7 & 2.8. Its value correct upto 10 places of decimal is 2.7182818284.

(b) Logarithms to the base 'e' are known as the Napierian system, so named after Napier, their inventor. They are also called **Natural Logarithm.**

(c) $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + ...\infty$; where x may be any real or complex number & $e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n$

(d) $a^x = 1 + \frac{x}{1!}\ell na + \frac{x^2}{2!}\ell n^2 a + \frac{x^3}{3!}\ell n^3 a + ...\infty$, where $a > 0$

(e) $e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + ...\infty$

2. Logarithmic Series

(a) $\ell n(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + ...\infty$, where $-1 < x \le 1$

(b) $\ell n(1 - x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} + ...\infty$, where $-1 \le x < 1$

Remember :

(i) $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + ...\infty = \ell n 2$ (ii) $e^{\ln x} = x$; for all $x > 0$

(iii) $\ell n2 = 0.693$ (iv) $\ell n10 = 2.303$

Q.1 The sum of the series $\sum_{n=1}^{\infty} \frac{n^2 + 6n + 10}{(2n+1)!}$ is equal to : **[O]**

श्रेणी $\sum_{n=1}^{\infty} \frac{n^2 + 6n + 10}{(2n+1)!}$ का योगफल बराबर है : **[O]**

Ans. $\frac{41}{8}e - \frac{19}{8}e^{-1} - 10$

Sol. $\sum_{n=1}^{\infty} \frac{n^2 + 6n + 10}{(2n+1)!}$

put $2n + 1 = r$

for $n = \{1, 2, 3, ...\}$

$r = \{3, 5, 7, ...\}$

$$= \sum_{r=3,5,7,....} \frac{\left(\frac{r-1}{2}\right)^2 + 6\cdot\left(\frac{r-1}{2}\right) + 10}{r!}$$

$$= \sum_{r=3,5,7,....} \frac{(r-1)^2 + 12(r-1) + 10}{4\lfloor r}$$

$$= \sum_{r=3,5,7,....} \frac{r^2 + 10r + 29}{4\lfloor r}$$

$$= \sum_{r=3,5,7,....} \frac{r(r-1)+11r+29}{4\lfloor r}$$

$$= \sum_{r=3,5,7,....} \left(\frac{1}{4(r-2)!} + \frac{11}{4(r-1)!} + \frac{29}{4} \cdot \frac{1}{r!} \right)$$

$$= \frac{1}{4}\left(\frac{1}{1!} + \frac{1}{3!} + \frac{1}{5!} + ... \right) + \frac{11}{4}\left(\frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + ... \right) + \frac{29}{4}\left(\frac{1}{3!} + \frac{1}{5!} + \frac{1}{7!} + ... \right)$$

$$= \frac{1}{4}\left(\frac{e - \frac{1}{e}}{2} \right) + \frac{11}{4}\left(\frac{e + \frac{1}{e} - 2}{2} \right) + \frac{29}{4}\left(\frac{e - \frac{1}{e} - 2}{2} \right)$$

$$= \frac{1}{8}\left[e - \frac{1}{e} + 11e + \frac{11}{e} - 22 + 29e - \frac{29}{e} - 58 \right]$$

$$= \frac{1}{8}\left[41e - \frac{19}{e} - 80 \right] = \frac{41}{8}e - \frac{19}{8}e^{-1} - 10$$